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Occultations by Small Non-spherical Trans-Neptunian Objects. I. A New Event Simulator for TAOS II

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Abstract

We present a new occultation event simulator for the Trans-Neptunian Automated Occultation Survey (TAOS II). We have developed a method to compute occultation shadows by small objects with non-circular apparent shapes (as may result from an intrinsic morphology or from the projection of a contact binary). The new simulator calculates diffraction features in the occultation shadows, as well as resulting light curves as would be measured by the TAOS II survey system. We include effects such as the spectral type and finite angular size of the occulted star. We find that occultation events, especially by Trans-Neptunian Objects with diameters $\sim$3 km may be misidentified or mischaracterized when not taking non-spherical shapes into account.

Key words: Kuiper belt: general – occultations – surveys

Online material: color figures

1. Introduction

The size distribution of Trans-Neptunian Objects (TNOs) has been measured for objects with diameters $D \gtrsim 25$ km (Bernstein et al. 2004; Fraser & Kavelaars 2008, 2009; Fuentes et al. 2009). The size distribution of TNOs with diameters $D > 90$ km can be reasonably modeled as a power law of the form:

$$\frac{dN}{dD} \propto D^{-q}, \tag{1}$$

with a slope $q \approx 4.5$ (Fraser et al. 2008; Fuentes & Holman 2008). The size distribution becomes shallower at $D < 90$ km (Shankman et al. 2013; Fraser et al. 2014; Alexandersen et al. 2016; Lawler et al. 2018), with a large uncertainty due to the relatively small number of detections and the fact that objects with $D \lesssim 50$ km are extremely difficult to detect. A measurement of the distribution for small TNOs ($0.5\,\mathrm{km} \lesssim D \lesssim 25$ km) is needed to further constrain the faint end of the TNO size distribution, because it would help constrain models of the dynamical evolution of the solar system (Kenyon & Bromley 2004, 2009; Pan & Sari 2005; Benavidez & Campo Bagatin 2009) as well as provide important information on the origin of short-period comets (Duncan & Levison 1997; Morbidelli 1997; Tancredi et al. 2006; Volk & Malhotra 2008).

Stellar occultations have become an important tool for the characterization of faint objects of the solar system (Elliot 1979; Leiva et al. 2017; Camargo et al. 2018; Gröller et al. 2018; Kammer et al. 2018; Kondratyev & Kornoukhov 2018; Pass et al. 2018; Arimatsu et al. 2019; Sickafoose et al. 2019). Bailey (1976) first proposed using a blind occultation survey to detect kilometer-sized TNOs, and Roques et al. (1987) first pointed out the significant diffraction features that would be present in any such occultation events. Additional theoretical considerations related to the feasibility of the technique were studied by Roques & Moncuquet (2000), Cooray & Farmer (2003), and Cooray (2003). The potential of occultation surveys for measuring the size distribution of small TNOs is such that several campaigns have been conducted, or will soon begin, using space and/or ground based telescopes (Roques et al. 2003, 2006; Bickerton et al. 2008; Bianco et al. 2009; Wang et al. 2010; Doressoundiram et al. 2013; Zhang et al. 2013; Lacki 2014; Subasinghe & Wiegert 2014; Liu et al. 2015; Santos-Sanz et al. 2016; Arimatsu et al. 2017). In addition, several surveys have reported detections consistent with occultation events by TNOs (Schlichting et al. 2009, 2012; Chang et al. 2011; Liu et al. 2015; Arimatsu et al. 2019). However, even if all of these claims of detections are correct, many more events will be needed in order to accurately constrain the size distribution of faint TNOs.

A next generation occultation survey will begin with the TAOS II project, which is expected to increase the effective sky coverage by a factor of 100 with respect to its predecessor (Lehner et al. 2012, 2014, 2016, 2018). This is achieved by increasing telescope aperture, locating at a site with better astroclimate, and imaging at a 20 Hz cadence using CMOS based cameras with low noise and multiple sub-frame readout around as many as 10,000 target stars simultaneously (Geary et al. 2012; Wang et al. 2014a, 2014b, 2016).

While TNOs with diameters of a few hundred kilometers have been found to be mostly spheroidal, with 85% of bodies characterized by axis ratio less than 1.5 (Luu & Lacerda 2003), it is not expected that this result will hold for much smaller, kilometer-size bodies, as these are certainly not massive enough to achieve hydrostatic equilibrium (Subasinghe & Wiegert 2014). Two studies have reported a correlation between light curve amplitude and absolute magnitude of small TNOs down to magnitude $H = 10.8$ (Benecchi & Sheppard 2013; Alexandersen et al. 2019), indicating that TNOs become more non-spherical as the sizes decrease. It would be reasonable to expect this trend to continue down to the kilometer-sized objects targeted by TAOS II. As a matter of fact, small objects that have been studied recently, are highly non-spherical such as: 2014 MU69 (Buie et al. 2018), 67P/ChuryumovGerasimenko (Cibulková et al. 2016) and "Oumuamua" (Vavilov & Medvedev 2019).

We have therefore created an occultation event simulator to study the "effects of non-spherical" TNOs. We include the effects of geocentric object distance, object size, angular size of occulted star, target star spectrum, filter transmission and detector quantum efficiency, reading direction, impact parameter, integration time and readout cadence in the generation of simulated light curves as would be measured by the TAOS II project. This simulator can be used to account for effects of non-spherical target shapes in the survey sensitivity. Furthermore, the simulator will provide a framework in which to study our light curve analysis to ensure that occultations by non-spherical TNOs are not needlessly rejected by our selection criteria, and to investigate the effect of shape on estimates of object size and distance for any detected events.

The paper is organized as follows. In Section 2 we review the simulation of the light curves resulting from the occultation by spherical TNOs. Section 3 details our methodology for simulating stellar occultation events by non-spherical objects, including the generation of the two-dimensional occultation shadows, and the generation of light curves as would be measured using the TAOS II telescopes and cameras. Section 4 shows comparisons between occultation events by spherical and non-spherical objects, and our conclusions are summarized in Section 5.

2. Occultations by Spherical Objects

Principles of diffraction have been extensively discussed by Cowley (1975), Born & Wolf (1987), Steward (1989), Goodman (1996), Saleh & Teich (2007) among others. In order to calculate the diffraction pattern caused by a small aperture or obstruction, there are two useful approximations depending on the Fresnel number:

$$F_{\mathrm{n}} = \frac{a^2}{z\lambda}, \tag{2}$$

where a is the size of the obstructing object, λ is the wavelength and z is the distance to the observer. If $F_{\mathrm{n}} \ll 1$ the Fraunhofer approach can be used, otherwise the Fresnel approach has to be computed. For an object of 2 km, at a distance of 40 au from Earth (Kuiper Belt) and a wavelength of 600 nm, the Fresnel number $F_{\mathrm{n}} \approx 1.11$, which is outside of the Fraunhofer diffraction regime. Hence, the Fresnel approach, which in general is reliable for any F_{n}, albeit more difficult to solve, must be used. The Fresnel approximation to the calculation of the amplitude of an electric field at a distance z from the obstruction to a plane wave is given by (Mas et al. 2000)

$$\begin{aligned} E_z(x_z, y_z) = E_0 \exp\left(i\frac{\pi}{\lambda z}(x_z^2 + y_z^2)\right) \\ \times \oint_S dx_0 dy_0 \left[\exp\left(i\frac{\pi}{\lambda z}(x_0^2 + y_0^2)\right)\right. \\ \left.\times \exp\left(-i\frac{2\pi}{\lambda z}(x_z x_0 + y_z y_0)\right)\right], \end{aligned} \tag{3}$$

where E_0 is the amplitude of the electric field at the obstruction, x_z, and y_z are the position of an arbitrary point in the observer plane, x_0 and y_0 are the points in the obscuration in the object plane, and the integral is over the surface of the obscuration S (see Figure 1). The measured intensity at a point in the shadow is simply

$$I_z(x_x, y_z) = |E_z(x_z, y_z)|^2. \tag{4}$$

Equation (3) is suitable to be solved analytically. However, the exact solution is constrained to simple geometrical shapes. For occultation surveys for TNOs, the circular shape is the

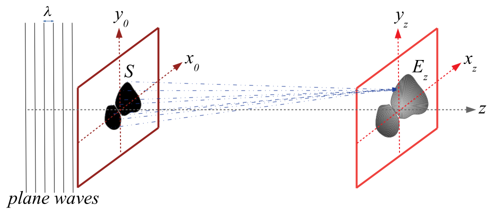

Figure 1. Scheme of the diffraction process according to Huygens–Fresnel principle. Note that we have used the Fresnel approximation, where the source is placed at infinity so it is possible to use plane waves, according to Equation (3).
(A color version of this figure is available in the online journal.)

most commonly used approximation, an assumption mostly used for simplicity as resulting from the 2D projection of a spherical shape for the TNOs in 3D. For circular geometry the solution to Equation (3), expressed in terms of Lommel functions, L_0, L_1 and L_2, is given by Roques et al. (1987) as

$$I(\rho, \eta) = \begin{cases} L_0^2(\eta, \rho) + L_1^2(\eta, \rho), & \eta \leqslant \rho, \\ 1 + L_1^2(\rho, \eta) + L_2^2(\rho, \eta) & \\ \quad -2L_1(\rho, \eta)\sin\dfrac{\pi}{2} & \eta \geqslant \rho, \\ +2L_2(\rho, \eta)\cos\dfrac{\pi}{2}(\rho^2 + \eta^2) & \end{cases} \quad (5)$$

where $\rho = r/F_s$ is the radius of the circular object in units of the Fresnel scale F_s, $\eta = x/F_s$ is the radial distance on the plane of the 2D occulting object, again in units of F_s, which is given by

$$F_s = \sqrt{\frac{\lambda z}{2}}. \quad (6)$$

From the solution shown in Equation (5), it is possible to derive all physical contributions to the diffraction profile and light curves as measured by an observer (Nihei et al. 2007).

3. Non-spherical Occultation Events

In this section we describe the procedures used to compute the diffraction pattern resulting from the occultation by an irregularly shaped body. In addition, as in Nihei et al. (2007), we proceed to calculate simulated light curves after including the effects such as the source star spectrum and angular size, impact parameter, opposition angle and time offset between event center and image cadence.

An earlier approach to understand and predict the influence of the object shape over the diffraction pattern was demonstrated by Roques et al. (1987). Starting with the Fresnel-Kirchhoff diffraction theory, they developed a method for computing the diffraction pattern of elliptical objects by treating them as a combination of long rectangles. A diffraction pattern is calculated by summing up the contribution to the electric field from each rectangle. Their tests did show significant differences in the light curves if the shape of the occulting object is non-spherical. For our event simulator we have developed a different approach using Fourier optics, which allows us to work in 2D with no restrictions in shape, and is very efficient due to the development of the Fast Fourier Transform algorithm (FFT).

3.1. The FFT Approach

In order to calculate the diffraction pattern produced by an opaque object of arbitrary shape, it is necessary to solve Equation (3) numerically. When $F_n \gtrsim 1$, this equation reduces itself to the two-dimensional Fourier Transformation (FT):

$$E_z(x_z, y_z) \propto \mathcal{F}(S) \quad (7)$$

of the projected surface of the object. However, for the Fresnel approach (near field) the complex field of the wavefront varies rapidly and the quadratic phase factors become very significant, so the expression becomes

$$E_z(x_z, y_z) \propto \mathcal{F}^{-1}\left[\mathcal{F}(S) \times \exp\left(\frac{-i\pi\lambda z}{(x_0^2 + y_0^2)}\right)\right], \quad (8)$$

that is, taking the FT of the object, multiplying by the quadratic factor, and then taking the inverse FT (Mas et al. 1999).

Equation (8) needs to be solved numerically. This technique has already been successfully implemented for calculating the influence of atmospheres in occultations by Young (2012). In our case, we substituted the FT by the FFT, since it is widely used and easy to implement.

While dealing with a well-known algorithm such as the FFT is convenient, discretization imposes an important drawback, that of sampling. In Fourier optics, the sampling resolution of the object and the corresponding spectrum are not independent, so, one must take care of the scaling produced by the sampling itself, especially when dealing with different object sizes. (Larger objects need lower sampling but their diffraction patterns need much higher sampling than is necessary for patterns of smaller objects, even though the diffraction shadow itself can be significantly larger.) If the object is over-sampled the spectrum will be under-sampled and vice versa. There are several methods to work around this issue (García et al. 1996; Mas et al. 1999; Hennelly & Sheridan 2005), most of them based on dynamically changing of the number of samples depending on the distance of propagation.

The sampling periods define the resolutions with which the objects are sampled. A high resolution corresponds to a small sampling period, and vice versa. In order to avoid the scaling problem just described, the number of points and the size of the object plane must be carefully chosen. Fortunately, the exact solution for circular objects given by Equation (5) is not affected by the scaling. Taking advantage of this fact, we use Equation (5) to compute a reference diffraction pattern, and by

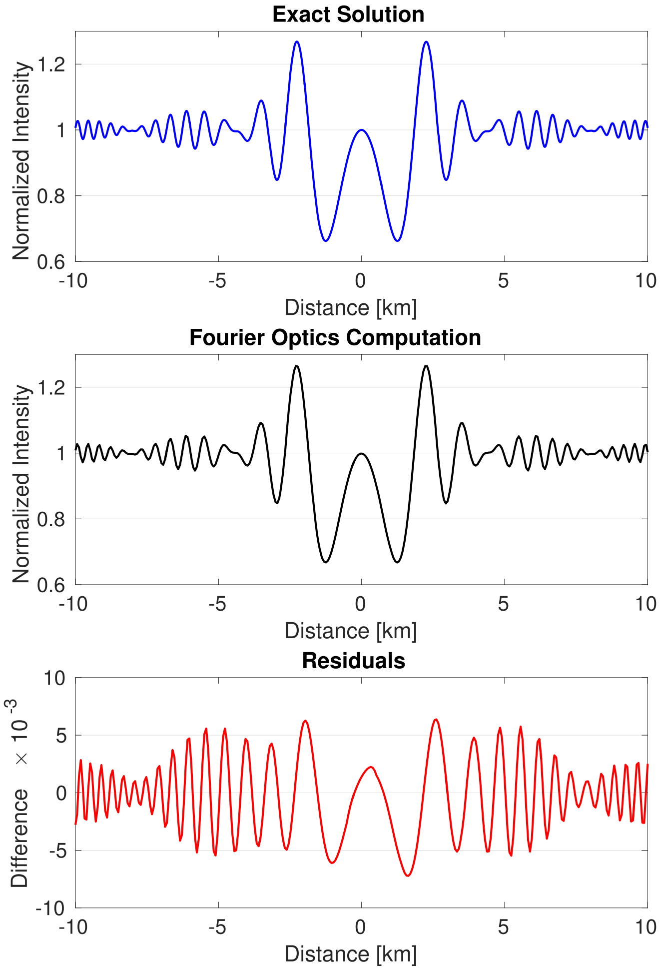

Figure 2. Diffraction profiles for a spherical object of 1 km diameter located at 40 au. Curves are taken for an observer crossing the center of the pattern, at a wavelength of 600 nm. In the top panel we show the exact solution calculated with Equation (5); in the middle, the pattern computed using the FFT (see Equation (8)) and in the bottom, the corresponding residuals. The residuals are typically <0.5%.
(A color version of this figure is available in the online journal.)

comparing it with that computed with Equation (8) for a circular object, it is possible to optimize the scale. Multiple calculations and comparisons have allowed us to empirically find that, for TNOs with $D = 20$ km, the size of the area in the object plane to be sampled when calculating the FFT must be about $50 \times F_s$ to get a good match with the exact solution. This sampling size turns out to be sufficient for smaller objects as well. This scaling criterion is thus completely suitable for the context of the TAOS II project, where the detection of small TNOs is the main objective. Figure 2 shows example diffraction profiles calculated for the same circular object using Equations (5) and (8), as well as the residuals between the two, which are typically <0.5%.

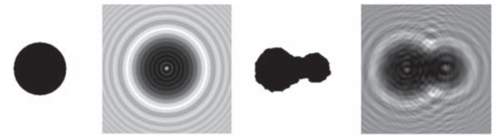

Figure 3. Illustration of the process for computing a (monochromatic) diffraction pattern for an object of 6.7 km at a distance of 43 au. From left to right: define a circular object, compute its diffraction pattern with exact solution to get the scale, define a new shape with the same area as the circular object, compute the new shape diffraction pattern using the FFT approach.

3.2. Generation of Diffraction Patterns

Throughout the remainder of this paper, we define the *effective diameter* of an object as

$$D_{\text{eff}} = \left(\frac{4A}{\pi}\right)^{\frac{1}{2}}, \tag{9}$$

where A is the projected area of the object. This simply means that two objects with the same effective diameter have the same projected area. (Note that the effective diameter of a non-spherical object thus depends on the orientation relative to the observer, as the projected area will change with rotation.) We define this quantity in order to have a point of comparison between spherical and non-spherical occultation events. This is the most obvious definition of "object size" for non-spherical objects since objects with the same D_{eff} will block the same number of photons from the occultation shadow.

In order to calculate the diffraction pattern for any arbitrary shape (in 2D), we proceed as follows:

1. Given an object size and a distance, we calculate the maximum size of the 2D pattern with Equation (6) and define the size of the plane of the object and the plane of the image according to the scale $50 \times F_s$.
2. Definition of the shape of the object S, keeping the same effective diameter as the reference spherical object.
3. Compute the diffraction pattern intensity $I(x, y)$ for the given shape of the object by using Equations (4) and (8).

This process is depicted schematically in Figure 3, corresponding to a TNO located at 43 au and $D_{\text{eff}} = 6.7$ km. Profiles of the diffraction patterns taken along a chord through the shadows (what a single telescope would observe as the diffracted shadow passes over it) for the circular and non-circular shapes are shown in Figure 4, illustrating the significant differences in the brightness patterns.

As a particular example, we consider an occultation by the comet 67P/ChuryumovGerasimenko, or 67P for short, as if it were observed when located at 43 au from Earth. The resulting diffraction pattern and a corresponding brightness profile

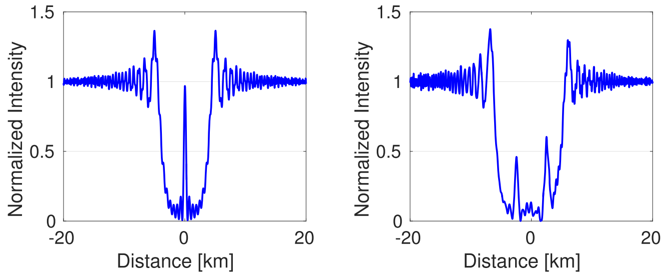

Figure 4. Monochromatic diffraction profiles along the x-axis through the center of mass of the two objects in Figure 3.
(A color version of this figure is available in the online journal.)

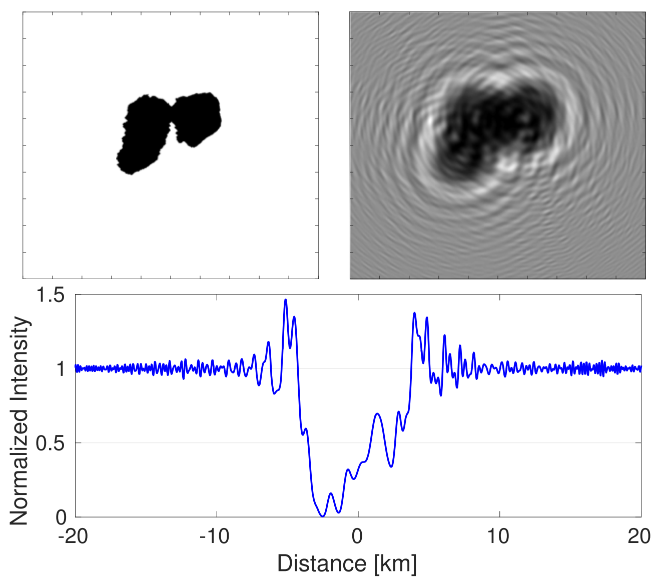

Figure 5. Monochromatic diffraction profile for comet 67P if it were at 43 au and instantaneously presenting the projected 2D shape shown in the top left panel. The top right panel shows the full diffraction pattern and the bottom panel shows a brightness profile along the x-axis at the middle of the pattern.
(A color version of this figure is available in the online journal.)

through the center of the object along the x-axis are shown in Figure 5.

3.3. Generation of Light Curves

We now follow the formalism presented in Nihei et al. (2007) to go from a diffraction pattern to a light curve as would be measured by an observatory system. Several modifications were needed for this process due to the loss of axial symmetry of the diffraction pattern with non-circular shapes.

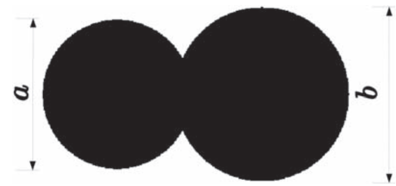

Figure 6. Example of compact binary used throughout the rest of the paper. The object comprises two spherical objects with diameters a and b in a ratio of $a{:}b = 0.85$. The centers of the objects are separated by $0.45 \times (a + b)$ in order to have some overlap.

For the remainder of this discussion, we define a single example non-spherical TNO as a contact binary (CB) comprising two spherical objects as shown in Figure 6. We keep the relative sizes and separations of the objects fixed as we vary $D_{\rm eff}$ in order to compare occultation events by spherical and non-spherical objects of different sizes.

3.3.1. Source Spectrum, Filter, and Detector Quantum Efficiency

Thus far we have assumed monochromatic diffraction patterns. However, real measurements of actual stars will cover a range of wavelengths with varying intensities, depending on the spectrum of the source star and the response of the observing system. In this work the stellar spectra were accounted for using the UVILIB library, a compiled stellar flux library spanning a total wavelength range of 115–1062 nm (Pickles 1998). We thus calculate the two-dimensional diffraction patterns averaged over wavelengths as

$$I(x, y) = \frac{1}{n}\sum_{k=1}^{n} I(x, y, \lambda_k) \times R_*(\lambda_k), \tag{10}$$

where $I(x, y, \lambda_k)$ is the monochromatic diffraction pattern calculated at wavelength λ_k, and $R_*(\lambda_k)$ is the weighting of the wavelength λ_k given by

$$R_*(\lambda_k) = s(\lambda_k) f(\lambda_k) q(\lambda_k). \tag{11}$$

Here $s(\lambda_k)$ is the normalized spectral intensity of the star at wavelength λ_k, and $f(\lambda_k)$ and $q(\lambda_k)$ are the filter transmittance and detector quantum efficiency at the same wavelength. The sum over k covers the full range of λ_k within the sensitivity of the observing system. In our simulation, the values of s, and f and q are stored in tables, with different tables for different stellar spectral types.

The effect of including these effects are illustrated in Figure 7. The consideration of the wide passband has the effect of smoothing out most of the high frequency variations of the light curve, but it has little effect on the main features of the event.

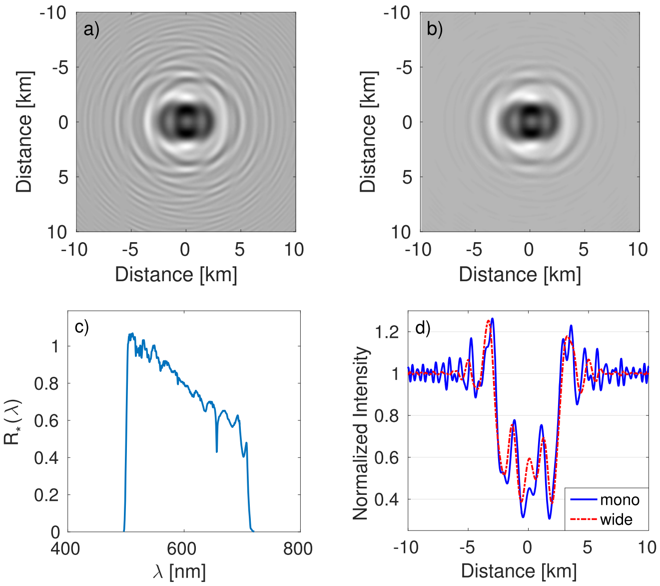

Figure 7. (a) Monchromatic diffraction pattern for the compact binary object shown in Figure 6, assuming $D_{\rm eff} = 2.5$ km, a distance of 50 au and a F0V target star. (b) Diffraction pattern calculated with Equation (10) using $R_*(\lambda_k)$ shown on the bottom left plot. (c) The response $R_*(\lambda_k)$ expected for a F0V star imaged with the TAOS II system. (d) Diffraction profiles along the x-axis through the object center for the monochromatic and full spectrum diffraction patterns shown in the top two panels.
(A color version of this figure is available in the online journal.)

3.3.2. *Finite Angular Size of Target Star*

Thus far, we have computed the diffraction patterns for arbitrary shape objects by considering the occulted stars as point sources. However, the effects of the finite angular sizes of the target stars can be important for brighter and late type stars, making the occultation patterns wider and shallower, as has been shown for circular objects (Nihei et al. 2007).

Given the luminosity and temperature of a star, the angular size can be estimated. The point source diffraction pattern must be integrated over this finite angular size, treating each point on the star as an individual point source. In analogy to Nihei et al. (2007) for the effect of a finite source in the diffraction pattern, we have implemented a numerical integration procedure in 2D.

We have tested our integration software by comparing its results with those of Nihei et al. (2007), and we found an excellent agreement. Examples of diffraction patterns for several different target stars are presented in the top panel of Figure 8. Angular sizes cover a range of 0.013 mas (A0V) to 0.18 mas (M2V), corresponding to projected sizes of 0.38–5.43 km in the TNO object plane. Brightness profiles through the centers of the patterns along the x-axes for the different spectral types are shown on the bottom panel of Figure 8. The effects of angular source size are evident for a fixed and bright apparent magnitude of all these stars, as the diffraction features are diminished and the events get wider with increasing angular size.

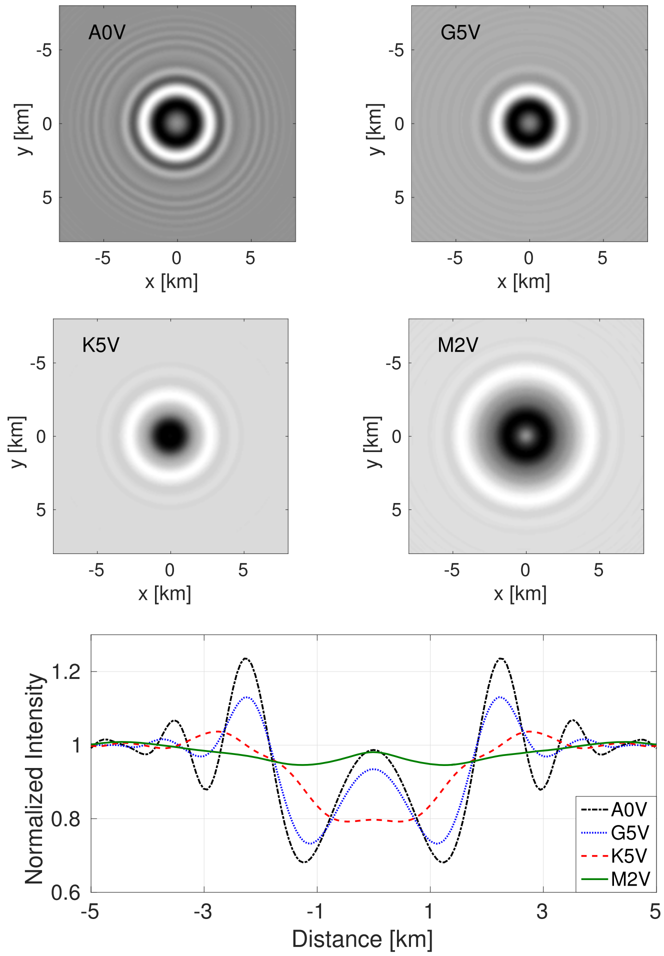

Figure 8. Top: Diffraction patterns for a circular 1 km diameter object at 40 au for several spectral types (A0V, G5V, K5V, and M2V), giving rise to a range of target star angular sizes. The apparent magnitude for each star is $V = 12$. Bottom: Diffraction profiles through the center of the diffraction patterns shown above.
(A color version of this figure is available in the online journal.)

3.3.3. *Impact Parameter and Reading Direction*

The impact parameter is simply defined as the minimum distance between the center of the occulting object and the line of sight from the observer to the target star (Nihei et al. 2007). In the case of a circular object, the intensity I_c of the occultation shadow along a chord at some impact parameter b is given by

$$I_c(x;\, b) = I[(x^2 + b^2)^{\frac{1}{2}}], \tag{12}$$

where x is the distance along the chord from the point of closest approach to the center of the occultation shadow, and $I(r)$ is the intensity at a radial distance r from the center of the (axially symmetric) diffraction pattern.

In the case of a non-circular shape, one additionally must consider the direction of motion of the shadow as well as the impact parameter. We thus define the *reading direction* as the angle of a chord across a diffraction pattern. The intensity along the chord thus becomes

$$I_{\rm nc}(x;\, b,\, \hat{\theta}) = I\,[x\cos(\hat{\theta}) - b\sin(\hat{\theta}),\, x\sin(\hat{\theta}) + b\cos(\hat{\theta})], \tag{13}$$

where $I(x,\, y)$ is the intensity of the diffraction pattern and $\hat{\theta}$ is the reading direction. For our simulation, we set the origin of the coordinate system to the center of mass of the projection of the object shape. The corresponding profiles for several values of b and $\hat{\theta}$ are shown in Figure 9. As expected, the impact parameter and reading direction combine to completely change the morphology of the diffraction profile.

3.3.4. Relative Velocity, Finite Exposure, Cadence, and Offset

The remaining steps to generate a simulated light curve are as follows:

1. Relative velocity. The relative velocity is dominated by the motion of the Earth around the Sun, and as such depends strongly on the opposition angle of the target star. Once the relative velocity is known, the conversion of distance x from the center of the event to time t is trivial.
2. Finite exposure. The diffraction profiles presented previously are shown with an infinitesimal exposure time. To calculate an actual light curve measurement, the diffraction profile (converted to the time domain in the previous step) must be integrated over the exposure time.
3. Cadence and offset. Any camera will have a minimum readout cadence which is at least as long as the exposure time. In addition, the center of the central epoch in the light curve will likely be offset from the time of closest approach of the center of the occultation shadow to the telescope.

These steps are identical to those taken for the generation of light curves for spherical objects, so we refer the reader to Nihei et al. (2007) for details.

4. Comparison of Light Curves from Spherical and Non-spherical TNOs

In this section we compare light curves of circular and CB shaped objects, for a range of effective diameters. In each case, we have used an object distance of 45 au, an A3V target star with apparent magnitude $V = 15$, and $\hat{\theta} = 0°$, and we assume observations at 30° from opposition, 50 ms exposures, a 20 Hz readout cadence, and no timing offsets. We compare light curves for objects with $D_{\rm eff} = 0.7$, 3.0, and 10.0 km, as these

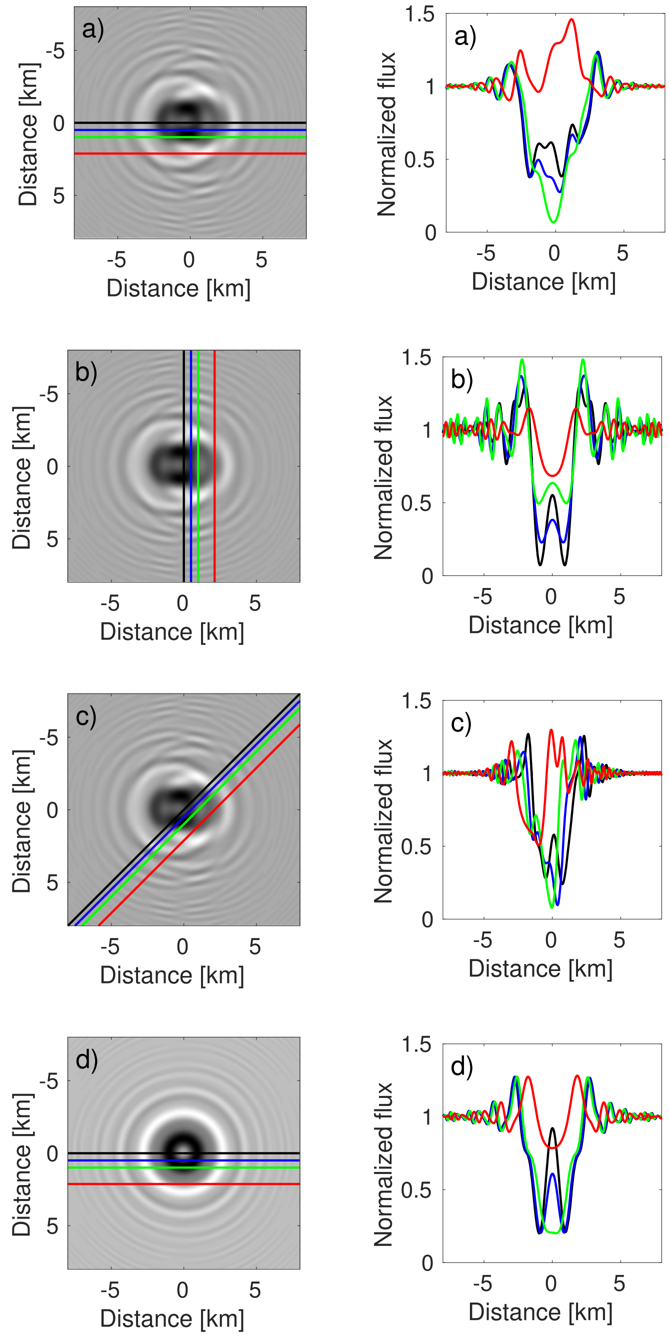

Figure 9. Diffraction patterns of a $D_{\rm eff} = 3$ km CB at 40 au occulting a A3V star, and resulting diffraction profiles at different values of b and $\hat{\theta}$. Each line color corresponds to a different impact parameter: $b = 0$ km (black), 0.7 km (blue), 1.4 km (green) and 3 km (red). Panels (a) to (c) show the results from the compact binary defined in Figure 6, while panels (d) show the results for a circular TNO for comparison. The reading directions for the CB are: (a) $\hat{\theta} = 0°$, (b) $\hat{\theta} = 90°$, and (c) $\hat{\theta} = 45°$.

(A color version of this figure is available in the online journal.)

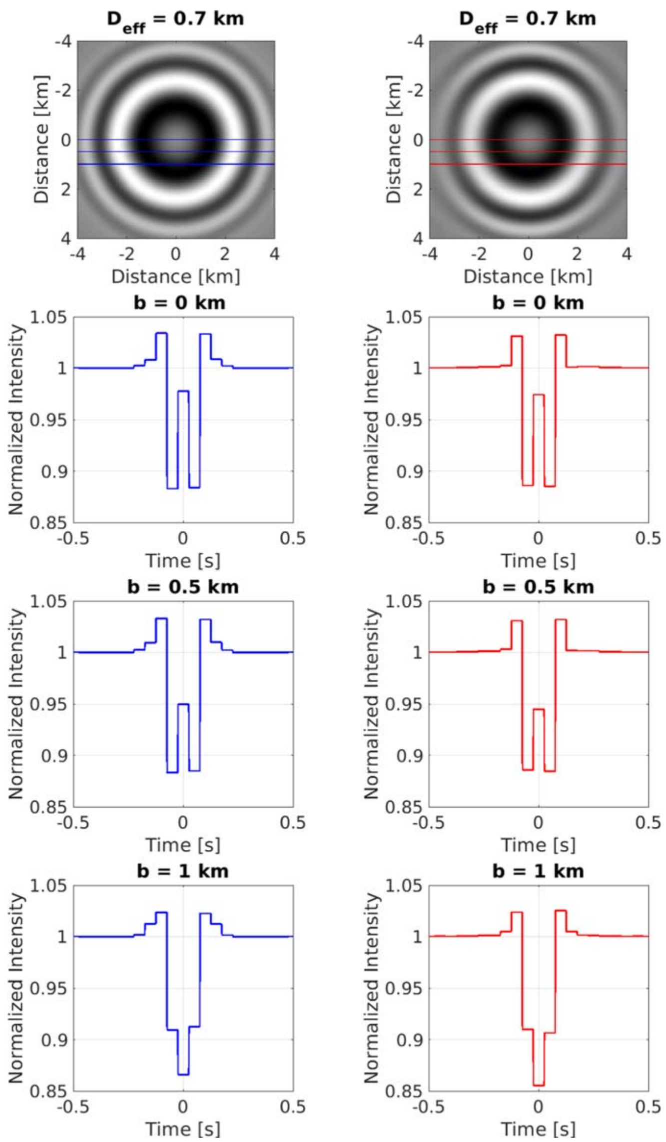

Figure 10. Comparison of light curves generated from the occultation of circular objects at the left column and CB objects at right column. The object size is $D_{\rm eff} = 0.7$ km. The impact parameter b was incremented in steps of 0.5 km. Other event parameters are described in the main text.

(A color version of this figure is available in the online journal.)

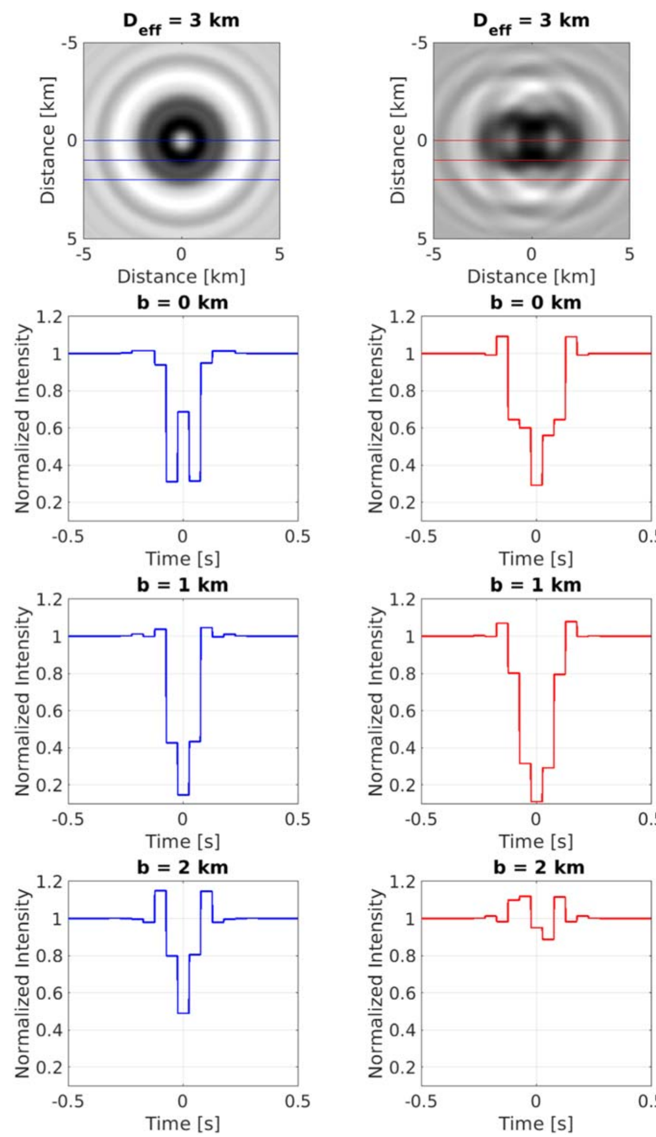

Figure 11. Comparison of light curves generated from the occultation of circular objects at the left column and CB objects at right column. The object size is $D_{\rm eff} = 3$ km. The impact parameter b was incremented in steps of 1 km. Other event parameters are described in the main text.

(A color version of this figure is available in the online journal.)

three sizes correspond to near, intermediate and far field in terms of diffraction regimes. In each case, we compare light curves at three values of impact parameter b. We include all of the effects discussed in Section 3.3.

In the far field regime ($D_{\rm eff} = 0.7$ km), there is little difference between the light curves from the spherical and non-spherical objects (see Figure 10). This is expected, as discussed in Goodman (2005).

In the intermediate regime ($D_{\rm eff} = 3.0$ km), there are significant differences between the light curve shapes (see Figure 11). The event appears wider at $b = 1$ km, and the non-spherical object event would be unlikely to be detectable at $b = 2$ km. While these conclusions will be different at different values of $\hat{\theta}$, it is clear that including the effects of non-spherical object shape will impact both the event detection efficiency and the event characterization.

In the near field regime ($D_{\rm eff} = 10$ km), the diffraction effects are minimal (see Figure 12). Nevertheless, the light curve shapes are different between the spherical and non-spherical cases. Indeed, different chords are being sampled across objects of different shapes, and the only parameter that can be measured in these cases are the lengths of the chords given an assumed distance. (Note that in the case of $b = 4$ km for the non-spherical object, the brightening in the middle of

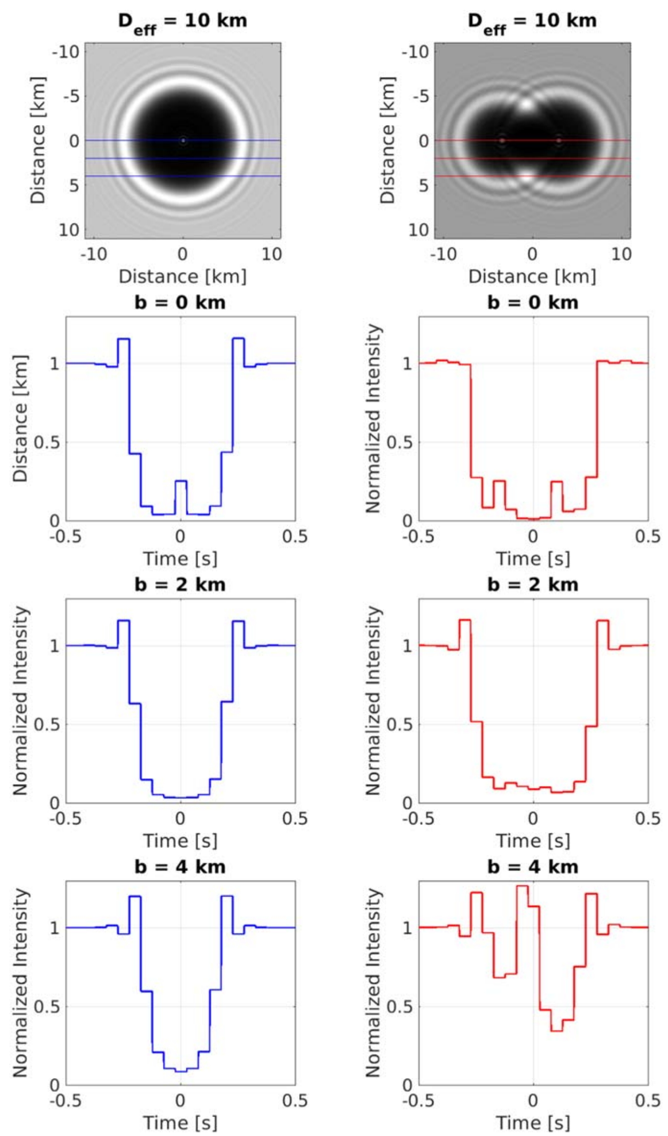

Figure 12. Comparison of light curves generated from the occultation of circular objects at the left column and CB objects at right column. The object size is $D_{\rm eff} = 10$ km. The impact parameter b was incremented in steps of 2 km. Other event parameters are described in the main text.

(A color version of this figure is available in the online journal.)

the light curve is due to the fact that the chord temporarily passes outside of the object.)

5. Conclusions

Non-circular shapes can result from intrinsic object morphology or from the projected shapes of binary objects. We expect most of the objects detected by TAOS II to be non-circular, and the investigation of the effects of object shape on the resulting occultation light curves is thus very important for determining location and size of the object.

We have presented the new simulator for TAOS II which can calculate occultation shadows from non-spherical TNOs. From these patterns we can derive occultation light curves as would be measured by an observatory system, taking into account the spectrum and angular size of the target star, filter transmission and detector sensitivity, impact parameter and reading direction, relative velocity between target and observer, and exposure and readout cadence. The algorithm is runs in the MATLAB environment, but we plan to implement an open source version of the simulator in the near future. Currently, the execution time to generate a single diffraction pattern using a single 4 GHz CPU is around 25 s. We expect the performance to improve significantly when the next version is finished.

We have presented comparisons of light curves from occultations by spherical and similarly sized non-spherical objects. Significant differences are evident between the light curves from the different object shapes with $D_{\rm eff} \gtrsim 1$ km.

For future work, we plan to develop and release an open source version of the light curve simulator, as discussed above. Furthermore, we plan to investigate the detection efficiency and accuracy of the object parameterization by running light curves from non-spherical TNOs through the first version of the TAOS II detection and event analysis pipeline. This investigation will inform the collaboration about the loss of detection efficiency and any biases and systematic uncertainties introduced in the estimation of sizes and distances of our detected events. We may modify our event detection criteria and/or our parameter estimation algorithms accordingly.

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Software: MATLAB (R2015b).

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